- **Tran Anh Chuong** - V202401566
- **Homework 1**

In [1]:
```
import pandas as pd
import matplotlib.pyplot as plt
```

In [2]:
```
df = pd.read_csv('ikea_data.csv')
df.head()
```

Out[2]:

	...1	item_id	name	category	price	old_price	sellable_online	link	other_colors
0	0	90420332	FREKVENS	Bar furniture	265.0	No old price	True	https://www.ikea.com/sa/en/p/frekvens-bar-tabl...	No
1	1	368814	NORDVIKEN	Bar furniture	995.0	No old price	False	https://www.ikea.com/sa/en/p/nordviken-bar-tab...	No
2	2	9333523	NORDVIKEN / NORDVIKEN	Bar furniture	2095.0	No old price	False	https://www.ikea.com/sa/en/p/nordviken-nordvik...	No
3	3	80155205	STIG	Bar furniture	69.0	No old price	True	https://www.ikea.com/sa/en/p/stig-bar-stool-wi...	Yes
4	4	30180504	NORBERG	Bar furniture	225.0	No old price	True	https://www.ikea.com/sa/en/p/norberg-wall-moun...	No

In [3]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3694 entries, 0 to 3693
Data columns (total 14 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   ...1               3694 non-null   int64
 1   item_id            3694 non-null   int64
 2   name               3694 non-null   object
 3   category           3694 non-null   object
 4   price              3694 non-null   float64
 5   old_price          3694 non-null   object
 6   sellable_online    3694 non-null   bool
 7   link               3694 non-null   object
 8   other_colors       3694 non-null   object
 9   short_description  3694 non-null   object
 10  designer           3551 non-null   object
 11  depth              2231 non-null   float64
 12  height             2706 non-null   float64
 13  width              3105 non-null   float64
dtypes: bool(1), float64(4), int64(2), object(7)
memory usage: 378.9+ KB
```

Task 1: Converting prices to USD

In [4]:
```
data = df.assign(price_usd = df['price']*0.27)
csv = data.loc[:, ['price','price_usd']]
csv.head(10)
```

Out[4]:

	price	price_usd
0	265.0	71.55
1	995.0	268.65
2	2095.0	565.65
3	69.0	18.63
4	225.0	60.75
5	345.0	93.15
6	129.0	34.83
7	195.0	52.65
8	129.0	34.83
9	2176.0	587.52

Task 2: Splitting multiple designers into separate rows

In [5]:

```
#Validate for rows contain multiple designers
data['designer'].unique()
```

Out[5]:
```
array(['Nicholai Wiig Hansen', 'Francis Cayouette', 'Henrik Preutz',
       'Marcus Arvonen', 'Carina Bengs', 'K Hagberg/M Hagberg',
       'Sarah Fager', 'Ehlén Johansson', 'Nike Karlsson',
       'Maja Ganszyniec', 'Karl Malmvall',
       'John/Jonas/Petrus/Paul/Caroline', 'Nike Karlsson/Maja Ganszyniec',
       'J Karlsson/N Karlsson', 'IKEA of Sweden/Karl Malmvall',
       'IKEA of Sweden', 'Nike Karlsson/J Karlsson/N Karlsson',
       'Ola Wihlborg', 'IKEA of Sweden/Tina Christensen',
       'IKEA of Sweden/K Hagberg/M Hagberg',
       'Ola Wihlborg/IKEA of Sweden', nan,
       'IKEA of Sweden/Ebba Strandmark',
       'K Hagberg/M Hagberg/IKEA of Sweden', 'Jon Karlsson',
       'IKEA of Sweden/Carina Bengs', 'David Wahl',
       'Jon Karlsson/IKEA of Sweden', 'IKEA of Sweden/Paulin Machado',
       'IKEA of Sweden/Eva Lilja Löwenhielm',
       'IKEA of Sweden/Synnöve Mork/Ola Wihlborg',
       'IKEA of Sweden/David Wahl', 'Ebba Strandmark/IKEA of Sweden',
       'Eva Lilja Löwenhielm', 'IKEA of Sweden/Anna Efverlund',
       'Paulin Machado', 'Jonas Hultqvist', 'Gustav Carlberg',
       'Carl Öjerstam', 'Virgil Abloh', 'Monika Mulder/IKEA of Sweden',
       'T Christensen/K Legaard', 'Tord Björklund',
       'Tom Dixon/IKEA of Sweden',
       'Synnöve Mork/Ola Wihlborg/IKEA of Sweden',
       'Ola Wihlborg/IKEA of Sweden/Synnöve Mork',
       'IKEA of Sweden/L Hilland/Lisa Hilland',
       'IKEA of Sweden/L Hilland', 'Carina Bengs/IKEA of Sweden',
       'Tom Dixon', 'Fredriksson/L Löwenhielm/Hilland',
       'IKEA of Sweden/Tord Björklund',
       'Fredriksson/L Löwenhielm/Hilland/IKEA of Sweden',
       'IKEA of Sweden/Henrik Preutz', 'IKEA of Sweden/Tom Dixon',
       'Malin Unnborn', 'A Huldén/S Dahlman',
       'IKEA of Sweden/Virgil Abloh', 'S Edholm/L Ullenius',
       'IKEA of Sweden/Francis Cayouette/Ehlén Johansson',
       'Francis Cayouette/Ehlén Johansson/IKEA of Sweden',
       'Ebba Strandmark', 'J Asshoff/H Brogård',
       'Lisa Hilland/David Wahl/IKEA of Sweden',
       'Henrik Preutz/IKEA of Sweden', 'C Halskov/H Dalsgaard',
       'Ehlén Johansson/IKEA of Sweden',
       'Ola Wihlborg/Synnöve Mork/IKEA of Sweden',
       'David Wahl/Lisa Hilland/IKEA of Sweden',
       'IKEA of Sweden/Ehlén Johansson/Francis Cayouette',
       'IKEA of Sweden/Fredriksson/L Löwenhielm/Hilland',
```

```
'IKEA of Sweden/Ehlén Johansson',
'Ehlén Johansson/Francis Cayouette/IKEA of Sweden', 'Chenyi Ke',
'Johan Kroon', 'Gillis Lundgren', 'IKEA of Sweden/Jon Karlsson',
'Gillis Lundgren/IKEA of Sweden', 'IKEA of Sweden/Carl Öjerstam',
'Sarah Fager/IKEA of Sweden',
'K Hagberg/M Hagberg/Gillis Lundgren',
'IKEA of Sweden/Marcus Arvonen', 'Marcus Arvonen/IKEA of Sweden',
'Gillis Lundgren/K Hagberg/M Hagberg', 'Niels Gammelgaard',
'H Preutz/A Fredriksson', 'IKEA of Sweden/Gillis Lundgren',
'Carl Öjerstam/IKEA of Sweden', 'Tord Björklund/IKEA of Sweden',
'Carl Öjerstam/Marcus Arvonen/IKEA of Sweden', 'Mikael Warnhammar',
'K Malmvall/E Lilja Löwenhielm',
'Gillis Lundgren/K Hagberg/M Hagberg/IKEA of Sweden',
'J Löfgren/J Pettersson',
'Gillis Lundgren/IKEA of Sweden/K Hagberg/M Hagberg',
'Charlie Styrbjörn', 'Tina Christensen',
'Francis Cayouette/Eva Lilja Löwenhielm',
'IKEA of Sweden/Sarah Fager', 'Francis Cayouette/IKEA of Sweden',
'Eva Lilja Löwenhielm/IKEA of Sweden',
'Marcus Arvonen/Carl Öjerstam/IKEA of Sweden',
'Eva Lilja Löwenhielm/K Malmvall/E Lilja Löwenhielm',
'Mia Lagerman', 'Lisa Norinder/Marcus Arvonen',
'Elizabet Gutierrez', 'Henrik Preutz/Olle Lundberg',
'Anna Palleschitz', 'A Fredriksson/J Hultqvist/W Chong',
'Lycke von Schantz', 'S Holmbäck/U Nordentoft', 'Jooyeon Lee',
'Lisa Hilland', 'Maria Vinka', 'Marcus Arvonen/Lisa Norinder',
'C Styrbjörn/M Axelsson',
'IKEA of Sweden/A Fredriksson/J Hultqvist/W Chong',
'Monika Mulder', 'Noboru Nakamura', 'Andreas Fredriksson',
'IKEA of Sweden/E Strandmark', 'M Kjelstrup/A Östgaard',
'K Hagberg/M Hagberg/Marcus Arvonen', 'Lisa Norinder',
'H Preutz/N Karlsson', 'Lars Norinder', 'E Thomasson/P Süssmann',
'David Wahl/Lisa Norinder', 'IKEA of Sweden/Noboru Nakamura',
'Anna Efverlund',
'Lisa Norinder/A Fredriksson/J Hultqvist/W Chong',
'Ola Wihlborg/Lisa Norinder',
'J Löfgren/J Pettersson/S Lanneskog/J Marnell',
'L Hilland/J Karlsson', 'Noboru Nakamura/IKEA of Sweden',
'Carl Öjerstam/S Lanneskog/J Marnell', 'Lisa Norinder/David Wahl',
'Mikael Axelsson/IKEA of Sweden',
'K Hagberg/M Hagberg/Mia Lagerman', 'IKEA of Sweden/Nike Karlsson',
'Jon Karlsson/John/Jonas/Petrus/Paul/Caroline',
```

'Nike Karlsson/Mikael Axelsson', 'Jon Karlsson/David Wahl',
'Olle Lundberg', 'HAY/A Fredriksson/J Hultqvist/W Chong',
'K Hagberg/M Hagberg/J Löfgren/J Pettersson',
'J Fritzdorf/J Feldman/J Hedberg', 'Ehlén Johansson/Ola Wihlborg',
'Ebba Strandmark/Carina Bengs',
'David Wahl/John/Jonas/Petrus/Paul/Caroline', 'S Fager/J Karlsson',
'Chris Martin', 'Ehlén Johansson/K Hagberg/M Hagberg',
'Marcus Arvonen/Nike Karlsson', 'P Süssmann/J Karlsson',
'IKEA of Sweden/S Lanneskog/J Marnell',
'Mia Lagerman/K Hagberg/M Hagberg',
'John/Jonas/Petrus/Paul/Caroline/Lisa Norinder',
'Ola Wihlborg/S Lanneskog/J Marnell',
'Carl Öjerstam/Ola Wihlborg/IKEA of Sweden',
'IKEA of Sweden/John/Jonas/Petrus/Paul/Caroline/David Wahl',
'Francis Cayouette/Maja Ganszyniec',
'Mia Lagerman/Mikael Warnhammar',
'Maja Ganszyniec/K Hagberg/M Hagberg',
'IKEA of Sweden/Carl Öjerstam/Mia Lagerman',
'S Lanneskog/J Marnell/IKEA of Sweden',
'Carl Öjerstam/Ehlén Johansson',
'Francis Cayouette/K Hagberg/M Hagberg',
'Carina Bengs/IKEA of Sweden/E Strandmark',
'S Lanneskog/J Marnell/Mia Lagerman',
'Mia Lagerman/S Lanneskog/J Marnell',
'S Lanneskog/J Marnell/Ola Wihlborg',
'J Karlsson/N Karlsson/Maja Ganszyniec',
'Mikael Warnhammar/Mia Lagerman',
'Maja Ganszyniec/J Karlsson/N Karlsson',
'Chris Martin/Ola Wihlborg/IKEA of Sweden',
'K Hagberg/M Hagberg/Francis Cayouette',
'David Wahl/IKEA of Sweden',
'IKEA of Sweden/E Strandmark/Carina Bengs',
'J Karlsson/N Karlsson/Nike Karlsson',
'Lisa Norinder/John/Jonas/Petrus/Paul/Caroline', 'HAY',
'Ehlén Johansson/J Löfgren/J Pettersson',
'Carina Bengs/Ebba Strandmark',
'Mia Lagerman/IKEA of Sweden/Wiebke Braasch',
'S Lanneskog/J Marnell', 'K Hagberg/M Hagberg/Maja Ganszyniec',
'Francis Cayouette/Mia Lagerman',
'K Hagberg/M Hagberg/John/Jonas/Petrus/Paul/Caroline',
'Mia Lagerman/Ehlén Johansson',
'Mikael Warnhammar/A Fredriksson/J Hultqvist/W Chong',

'S Lanneskog/J Marnell/J Löfgren/J Pettersson',
'Mia Lagerman/IKEA of Sweden/Chris Martin',
'David Wahl/IKEA of Sweden/John/Jonas/Petrus/Paul/Caroline',
'Karl Malmvall/Ehlén Johansson', 'Francis Cayouette/Nike Karlsson',
'J Löfgren/J Pettersson/Marcus Arvonen',
'Ola Wihlborg/Ehlén Johansson', 'Mia Lagerman/Francis Cayouette',
'Nike Karlsson/Francis Cayouette', 'IKEA of Sweden/Ola Wihlborg',
'Johanna Asshoff', 'T Winkel/T Jacobsen', 'Studio Copenhagen',
'Eva Schildt', 'S Fager/J Jelinek',
'Studio Copenhagen/Mia Lagerman', 'Thomas Sandell', 'Annie Huldén',
'IKEA of Sweden/Andreas Fredriksson',
'Eva Lilja Löwenhielm/K Hagberg/M Hagberg/IKEA of Sweden',
'Eva Lilja Löwenhielm/IKEA of Sweden/K Hagberg/M Hagberg',
'Jonas Hultqvist/IKEA of Sweden/Eva Lilja Löwenhielm',
'Marcus Arvonen/Andreas Fredriksson',
'IKEA of Sweden/Eva Lilja Löwenhielm/K Hagberg/M Hagberg',
'Magnus Elebäck',
'Eva Lilja Löwenhielm/IKEA of Sweden/Nike Karlsson',
'IKEA of Sweden/Jonas Hultqvist',
'Eva Lilja Löwenhielm/Jonas Hultqvist/IKEA of Sweden',
'Mikael Axelsson',
'Eva Lilja Löwenhielm/IKEA of Sweden/Jonas Hultqvist',
'Eva Lilja Löwenhielm/IKEA of Sweden/David Wahl',
'Lisel Garsveden',
'Jon Karlsson/IKEA of Sweden/Eva Lilja Löwenhielm',
'IKEA of Sweden/David Wahl/Eva Lilja Löwenhielm',
'Nike Karlsson/IKEA of Sweden',
'Nike Karlsson/IKEA of Sweden/Eva Lilja Löwenhielm',
'IKEA of Sweden/Nike Karlsson/Eva Lilja Löwenhielm',
'Jonas Hultqvist/IKEA of Sweden',
'IKEA of Sweden/David Wahl/Lisa Hilland', 'Synnöve Mork',
'Lisa Hilland/IKEA of Sweden/David Wahl',
'Francis Cayouette/Ehlén Johansson',
'Ehlén Johansson/Francis Cayouette',
'Ehlén Johansson/Fredriksson/L Löwenhielm/Hilland',
'IKEA of Sweden/Fredriksson/L Löwenhielm/Hilland/Ehlén Johansson',
'IKEA of Sweden/Francis Cayouette',
'Francis Cayouette/IKEA of Sweden/Ehlén Johansson', 'Nada Debs',
'Johanna Asshoff/IKEA of Sweden', 'IKEA of Sweden/Johanna Asshoff',
'IKEA of Sweden/Chenyi Ke/Johanna Asshoff', 'Johanna Jelinek',
'Johanna Asshoff/Jomi Evers', 'Gustav Carlberg/Johanna Asshoff',
'IKEA of Sweden/Gustav Carlberg', 'Jomi Evers',

```
       'Chenyi Ke/IKEA of Sweden', 'IKEA of Sweden/Jomi Evers',
       'Francis Cayouette/Jomi Evers', 'Gustav Carlberg/IKEA of Sweden',
       'Jomi Evers/IKEA of Sweden',
       'Johanna Asshoff/IKEA of Sweden/Gustav Carlberg',
       'Johanna Asshoff/Gustav Carlberg', 'Chris Martin/IKEA of Sweden',
       'IKEA of Sweden/Chenyi Ke', 'Wiebke Braasch',
       'Francis Cayouette/Gustav Carlberg', 'Tina Christensen/Jomi Evers',
       'IKEA of Sweden/Marcus Arvonen/Carl Öjerstam',
       'J Karlsson/W Chong',
       'IKEA of Sweden/K Hagberg/M Hagberg/Ehlén Johansson',
       'K Hagberg/M Hagberg/Ehlén Johansson/IKEA of Sweden',
       'IKEA of Sweden/Ehlén Johansson/K Hagberg/M Hagberg',
       'Ehlén Johansson/K Hagberg/M Hagberg/IKEA of Sweden',
       'IKEA of Sweden/Ebba Strandmark/Ehlén Johansson',
       'Ehlén Johansson/IKEA of Sweden/K Hagberg/M Hagberg',
       'IKEA of Sweden/Ehlén Johansson/Ola Wihlborg',
       'Ebba Strandmark/Ehlén Johansson/IKEA of Sweden',
       'Ebba Strandmark/IKEA of Sweden/Ola Wihlborg/Ehlén Johansson',
       'IKEA of Sweden/Ola Wihlborg/Ehlén Johansson',
       'Ehlén Johansson/Ebba Strandmark/IKEA of Sweden',
       'IKEA of Sweden/Ehlén Johansson/Ebba Strandmark',
       'IKEA of Sweden/Ola Wihlborg/Ehlén Johansson/Ebba Strandmark',
       'Ehlén Johansson/Ebba Strandmark/Ola Wihlborg/IKEA of Sweden',
       'IKEA of Sweden/Andreas Fredriksson/Ehlén Johansson',
       'Ehlén Johansson/Ola Wihlborg/IKEA of Sweden',
       'Ola Wihlborg/Ehlén Johansson/IKEA of Sweden',
       'Ehlén Johansson/Andreas Fredriksson/IKEA of Sweden',
       'IKEA of Sweden/Ehlén Johansson/Andreas Fredriksson'], dtype=object)
```

The designer column may contain multiple designers separated by /. ✅ Transform the dataset so that each designer appears in a separate row with the same item details

In [6]:
```python
# Step 1: Split the 'designer' string by '/' into a list of strings
data['designer'] = data['designer'].str.split('/')
# Step 2: Explode the 'designer' column so each item in the list gets its own row
data = data.explode('designer')
# Step 3: Strip any extra spaces around the designer names
data['designer'] = data['designer'].str.strip()
# Display the result to confirm it worked!
data.head(11)
```

Out[6]:

	...1	item_id	name	category	price	old_price	sellable_online	link	other_colors
0	0	90420332	FREKVENS	Bar furniture	265.0	No old price	True	https://www.ikea.com/sa/en/p/frekvens-bar-tabl...	No
1	1	368814	NORDVIKEN	Bar furniture	995.0	No old price	False	https://www.ikea.com/sa/en/p/nordviken-bar-tab...	No
2	2	9333523	NORDVIKEN / NORDVIKEN	Bar furniture	2095.0	No old price	False	https://www.ikea.com/sa/en/p/nordviken-nordvik...	No
3	3	80155205	STIG	Bar furniture	69.0	No old price	True	https://www.ikea.com/sa/en/p/stig-bar-stool-wi...	Yes
4	4	30180504	NORBERG	Bar furniture	225.0	No old price	True	https://www.ikea.com/sa/en/p/norberg-wall-moun...	No
5	5	10122647	INGOLF	Bar furniture	345.0	No old price	True	https://www.ikea.com/sa/en/p/ingolf-bar-stool-...	No
6	6	70404875	FRANKLIN	Bar furniture	129.0	No old price	True	https://www.ikea.com/sa/en/p/franklin-bar-stoo...	No
6	6	70404875	FRANKLIN	Bar furniture	129.0	No old price	True	https://www.ikea.com/sa/en/p/franklin-bar-stoo...	No
7	7	60155602	DALFRED	Bar furniture	195.0	No old price	True	https://www.ikea.com/sa/en/p/dalfred-bar-stool...	No
8	8	50406465	FRANKLIN	Bar furniture	129.0	No old price	True	https://www.ikea.com/sa/en/p/franklin-bar-stoo...	No
8	8	50406465	FRANKLIN	Bar furniture	129.0	No old price	True	https://www.ikea.com/sa/en/p/franklin-bar-stoo...	No

Task 3: Get Top 20 Designers by Number of Items

Based on the dataset from Task 2, find the top 20 designers with the most products. ✅ Exclude "IKEA of Sweden" and remove missing values from the designer column.

In [7]:
```
top_designer = data.groupby('designer').size().nlargest(5).reset_index(name='num_items')
top_designer.head()
```

Out[7]:

	designer	num_items
0	IKEA of Sweden	1502
1	Ehlén Johansson	394
2	Francis Cayouette	217
3	Ola Wihlborg	210
4	K Hagberg	195

- **1. Top 3 designers are Ehlen Johansson (394) and Francis Cayouette (217) and Ola Wihlborg (210)**
- **2. Number of items designed by IKEA of Sweden is 1502**
- **3. We should exclude IKEA of Sweden since it represents the organization instead of individual designer, thus we can avoid bias.**

In [8]:
```
top_designer = data.dropna(subset=['designer'])
top_designer = top_designer[top_designer['designer'] != 'IKEA of Sweden']
top_designer = top_designer.groupby('designer').size().nlargest(3).reset_index(name='num_items')
top_designer.head()
```

Out[8]:

	designer	num_items
0	Ehlén Johansson	394
1	Francis Cayouette	217
2	Ola Wihlborg	210

Task 4: Price Distribution per Designer (Box Plot)

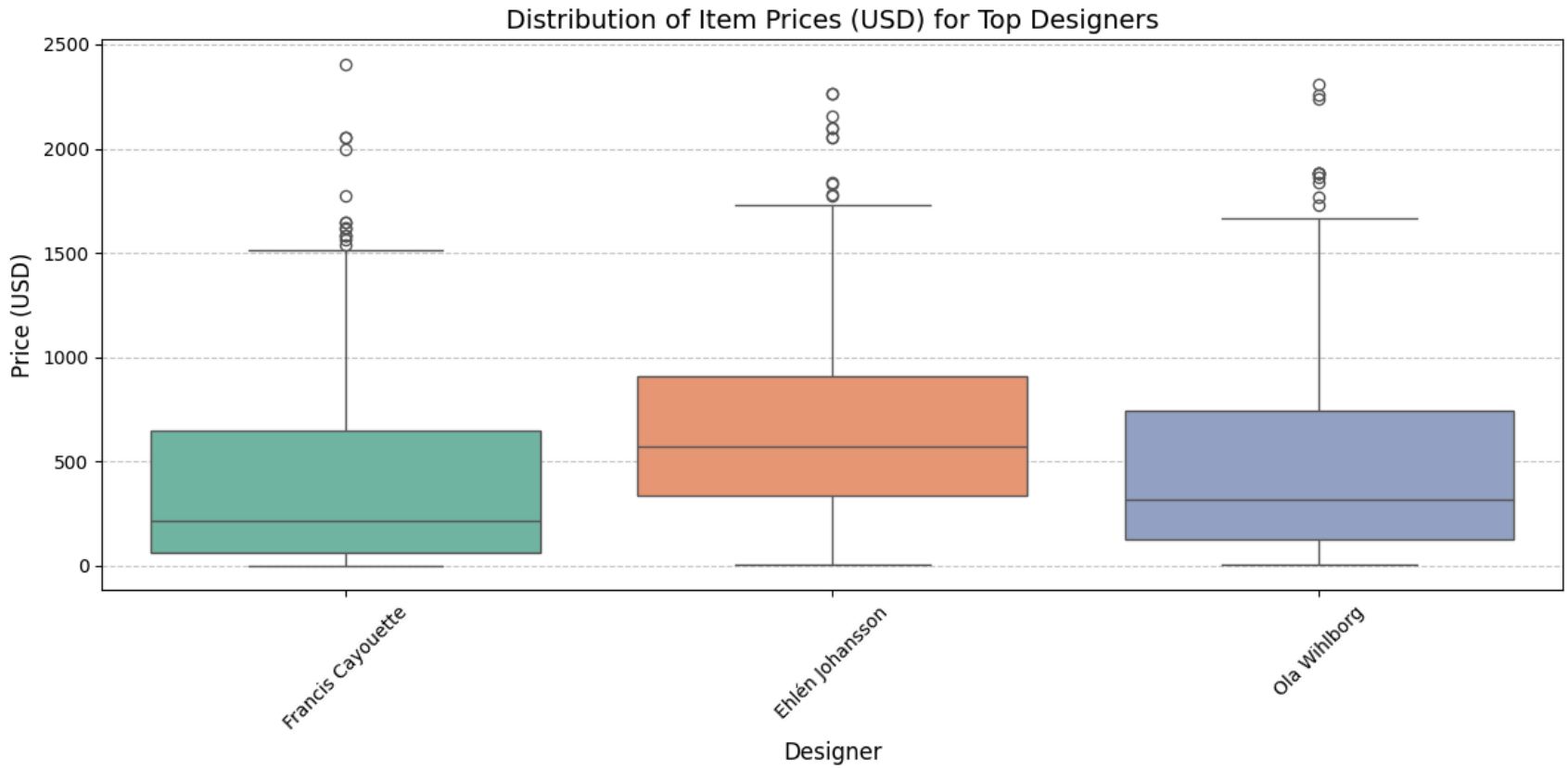
✅ Create a box plot showing the distribution of item prices (USD) for the top designers identified in Task 3.

In [9]:

```
import seaborn as sns
import matplotlib.pyplot as plt

# 1. We need the list of the top designers' names from Task 3
top_designer_names = top_designer['designer'].tolist()

# 2. Filter the main dataset (`data` from Task 1 which has 'price_usd')
# to only include rows where the designer is in our top designers list
top_designer_data = data[data['designer'].isin(top_designer_names)]

# 3. Create the box plot using Seaborn for beautiful default styling
plt.figure(figsize=(12, 6))

# We use seaborn's boxplot, mapping designer to x and price_usd to y
sns.boxplot(
    data=top_designer_data,
    x='designer',
    y='price_usd',
    palette='Set2'
)

# 4. Add labels and a title to make it professional
plt.title('Distribution of Item Prices (USD) for Top Designers', fontsize=14)
plt.xlabel('Designer', fontsize=12)
plt.ylabel('Price (USD)', fontsize=12)

# Rotate the x-axis labels so the designers' names don't overlap
```

```
plt.xticks(rotation=45)

# Show the plot cleanly
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.tight_layout()
plt.show()
```

```
C:\Users\jimmy\AppData\Local\Temp\ipykernel_27664\2868704391.py:15: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `h
ue` and set `legend=False` for the same effect.

  sns.boxplot(
```



- **Key findings:** Ehlen product's price is higher than the other two indicating this designer's level, median is roughly $600. Francis and Ola sharesimilardistributioninprice; Francis' sproductisthemostbudget - friendlywiththemedianof$

250. Ola has the greatest variety in the market, the median is around $300andtheboxishigherdepictingawiderspreadinthemiddle501500$ to around $2200. 3 designers have median price close to the bottom of the box.

- **For me this is a relatively effective visualization. I think we can add a median number for each boxes for a better quick overview for the viewers.**

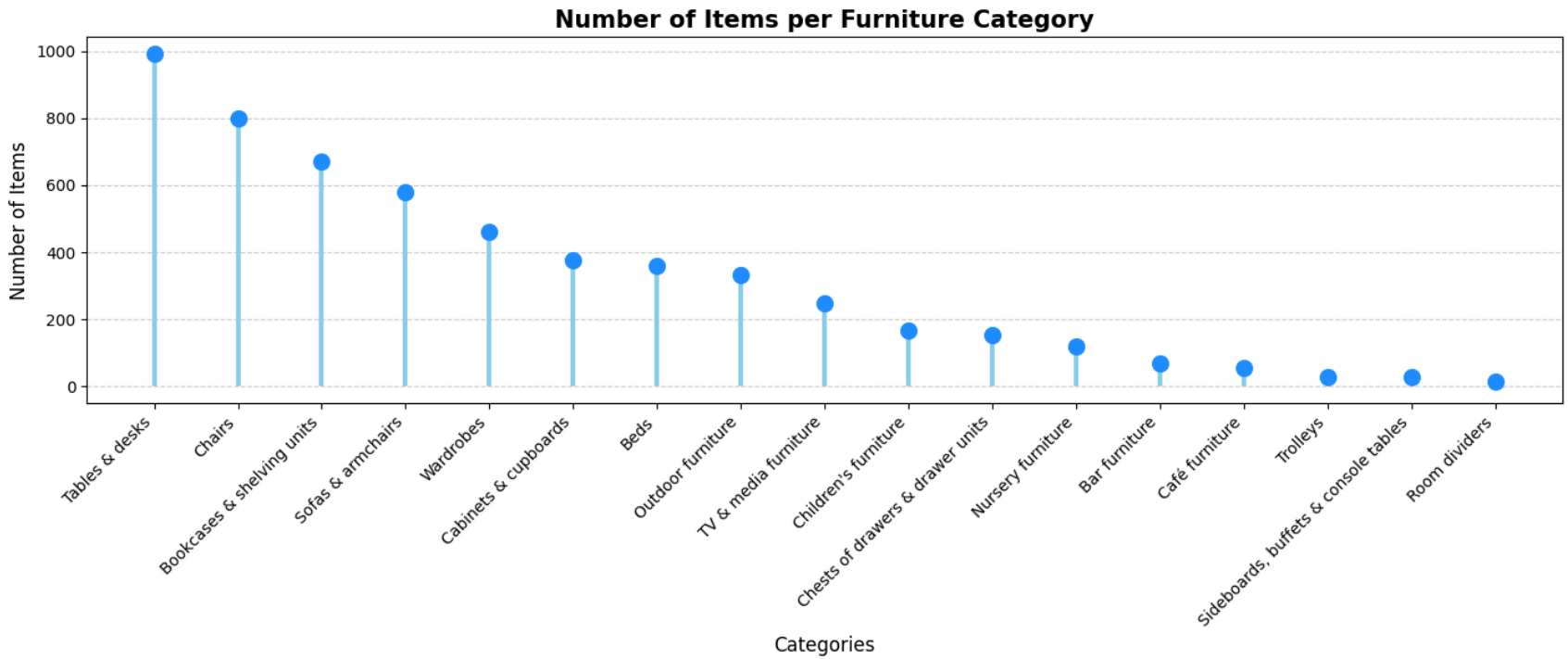
Task 5: Distribution of Items per Category (Lollipop Chart)

✅ Count the number of items in each furniture category and visualize the distribution using a lollipop chart.

- X-axis: Categories
- Y-axis: Number of items

In [10]:
```
# 1. Count the number of items in each furniture category
# reset_index() converts the resulting Series back into a DataFrame
category_counts = data['category'].value_counts().reset_index()
category_counts.columns = ['category', 'num_items']

# 2. Create the figure for the plot
plt.figure(figsize=(14, 6))

# 3. Create the Lollipop Chart
# A lollipop chart is essentially a vertical line connected to a scatter dot!
# The 'vlines' draw the stems
plt.vlines(
    x=category_counts['category'],
    ymin=0,
    ymax=category_counts['num_items'],
    color='skyblue',
    linewidth=3
)

# The 'scatter' draws the candy on top
plt.scatter(
    x=category_counts['category'],
    y=category_counts['num_items'],
    color='dodgerblue',
```

```python
    s=100, # Size of the dots
    zorder=3 # Ensures the dots render on top of gridlines/stems
)

# 4. Add formatting and labels
plt.title('Number of Items per Furniture Category', fontsize=15, fontweight='bold')
plt.xlabel('Categories', fontsize=12)
plt.ylabel('Number of Items', fontsize=12)

# Rotate the category names so they don't overlap
plt.xticks(rotation=45, ha='right')

# Add a light background grid for easy reading
plt.grid(axis='y', linestyle='--', alpha=0.6, zorder=0)

# Display the plot
plt.tight_layout()
plt.show()
```



- **Most common categories:** Tables & desks (around 1000) followed by Chairs (approx 800)
- **We can add exact number for each categories, change the diagram to horizontal view since the categories name are quite long (45 deg rotation can cause inconvinience)**

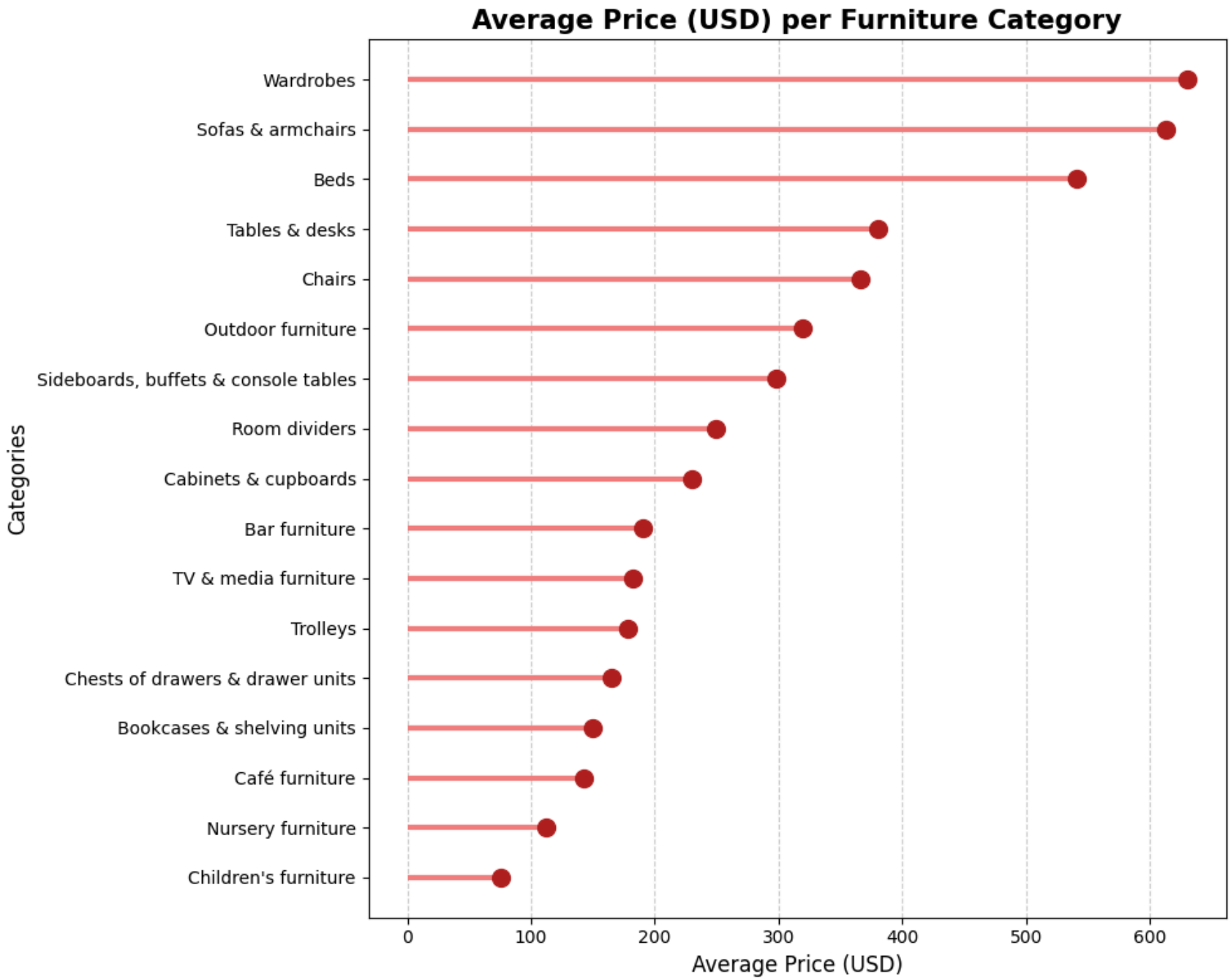
Task 6: Average Price per Category

✅ Calculate the average price (USD) per furniture category and visualize the distribution using a lollipop chart.

- X-axis: Categories
- Y-axis: Average price in USD

In [11]:
```
# 1. Calculate the average price (USD) per furniture category
# We group by 'category', find the mean of 'price_usd', and sort ascending
# so the highest value appears at the top of the horizontal chart!
avg_price_cat = data.groupby('category')['price_usd'].mean().sort_values(ascending=True).reset_index()

# 2. Create the figure for the plot
plt.figure(figsize=(10, 8))

# 3. Create the Horizontal Lollipop Chart
# plt.hlines() draws the horizontal stems
plt.hlines(
    y=avg_price_cat['category'],
    xmin=0,
    xmax=avg_price_cat['price_usd'],
    color='lightcoral',
    linewidth=3
)

# plt.scatter() draws the candy on the right side!
plt.scatter(
    x=avg_price_cat['price_usd'],
    y=avg_price_cat['category'],
    color='firebrick',
    s=100,
    zorder=3 # Ensures the dots render clearly on top of grid lines
)
```

```python
# 4. Add formatting and titles
plt.title('Average Price (USD) per Furniture Category', fontsize=15, fontweight='bold')
plt.xlabel('Average Price (USD)', fontsize=12)
plt.ylabel('Categories', fontsize=12)

# Add a clean background grid on the x-axis to make values easy to read
plt.grid(axis='x', linestyle='--', alpha=0.6, zorder=0)

# Render everything nicely
plt.tight_layout()
plt.show()
```


Average Price (USD) per Furniture Category
Wardrobes
Sofas & armchairs
Beds
Tables & desks
Chairs
Outdoor furniture
Sideboards, buffets & console tables
Room dividers
Cabinets & cupboards
Bar furniture
TV & media furniture
Trolleys
Chests of drawers & drawer units
Bookcases & shelving units
Café furniture
Nursery furniture
Children's furniture
Categories
0
100
200
300
400
500
600
Average Price (USD)

- **Most expensive furniture categories:** Wardrobes, Sofas & armchairs, Beds with the price of around $630,610 and $530 respectively.
- **Cheapest furniture categories:** Children's furniture (~$80), $Nurseryfurniture($ 105) and cafe furniture (~$140)
- **Suprising findings:** Although Tables & desks among the top 5 most expensive items, they are the most common items in the previous analysis. This suggests that while they are high-volume, they aren't necessarily "cheap" filler items.

Task 7: Price vs. Volume Relationship (Scatter Plot)

✅ Calculate the volume of each item using: Volume=Height×Width×Depth\text{Volume} = \text{Height} \times \text{Width} \times \text{Depth}Volume=Height×Width×Depth ✅ Create a scatter plot showing the relationship between price (USD) and volume.

- X-axis: Price (USD)
- Y-axis: Volume
- Color: Category

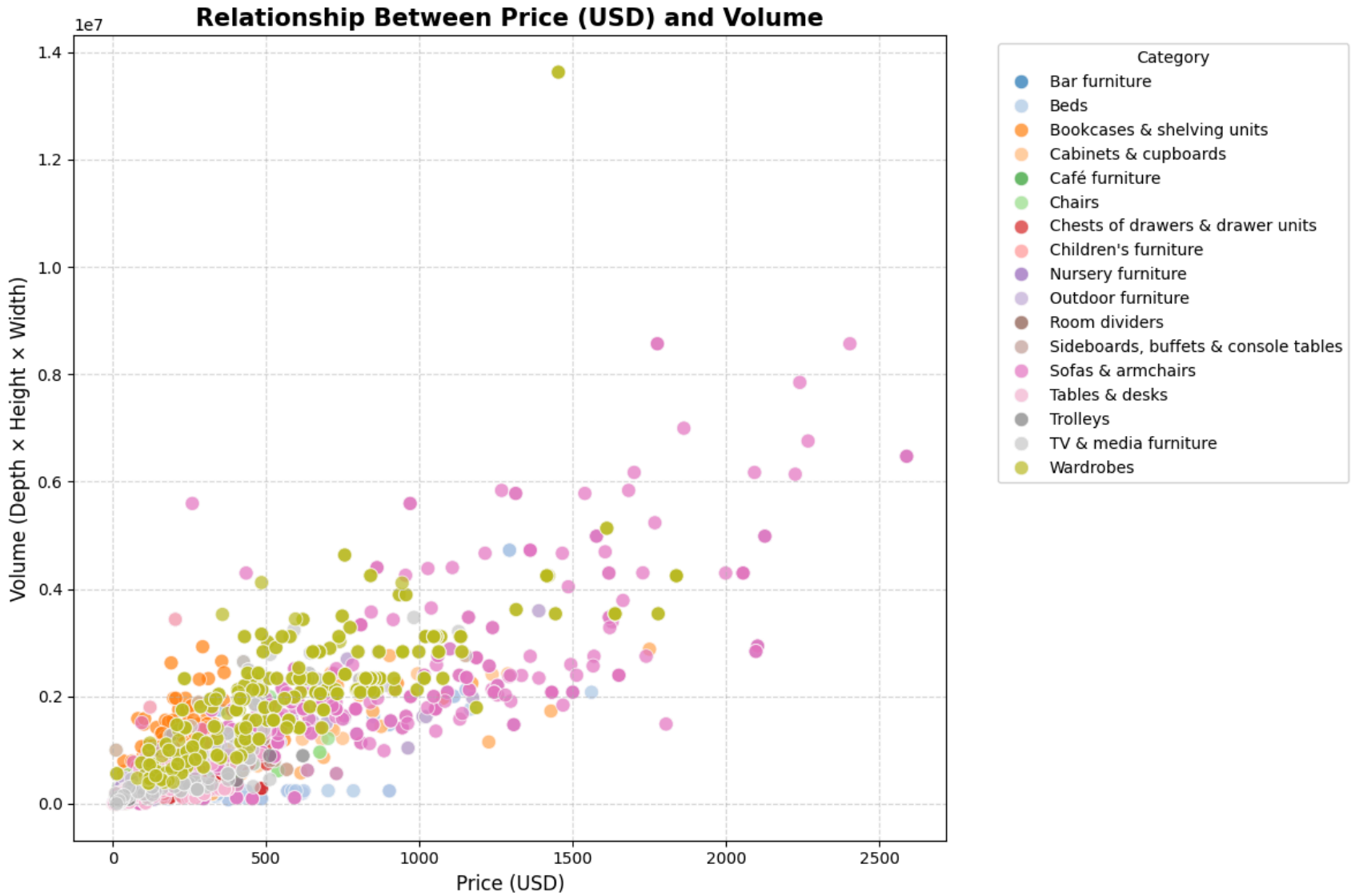
In [12]:
```
test = data.loc[:, ['height', 'width', 'depth']]
test.head()
```

Out[12]:

	height	width	depth
0	99.0	51.0	NaN
1	105.0	80.0	NaN
2	NaN	NaN	NaN
3	100.0	60.0	50.0
4	43.0	74.0	60.0

In [13]:
```
# 1. Reset the index to fix duplicate labels from Task 2, then calculate volume
data = data.reset_index(drop=True)
data['volume'] = data['depth'] * data['height'] * data['width']

# 2. Create the figure for the scatter plot
plt.figure(figsize=(12, 8))
```

```python
# 3. Create the Scatter Plot
# x='price_usd', y='volume', hue='category' colors it by category automatically
sns.scatterplot(
    data=data,
    x='price_usd',
    y='volume',
    hue='category',
    palette='tab20',   # A nice color palette with distinct colors
    alpha=0.7,         # Make dots slightly transparent so overlapping points are visible
    s=80               # Adjust dot size to be easier to see
)

# 4. Add titles and formatting
plt.title('Relationship Between Price (USD) and Volume', fontsize=15, fontweight='bold')
plt.xlabel('Price (USD)', fontsize=12)
plt.ylabel('Volume (Depth × Height × Width)', fontsize=12)

# Position the legend outside the plot so it doesn't cover up our data
plt.legend(bbox_to_anchor=(1.05, 1), loc='upper left', title='Category')

# 5. Bring in the visual polish
plt.grid(True, linestyle='--', alpha=0.5)
plt.tight_layout() # This forces the plot to fit nicely considering the external legend
plt.show()
```



- **1. Are larger items always more expensive?** No, larger items are not always more expensive. While there is a general upward trend—meaning that as volume increases, price tends to increase—the relationship is quite "noisy" and far from a perfect linear correlation. Materials and Complexity: A small item made of premium materials (like a designer chair or a high-end electronics console) can easily outprice a massive but simple item (like a basic particle-board wardrobe). The "Luxury Small" items: If you

look at the bottom-right of the plot, there are several dots representing relatively small items (low volume) that cost between $1,500 and 2,500. The "Massive Value" items: Conversely, toward the middle-top, there are very large items priced under $1,000.

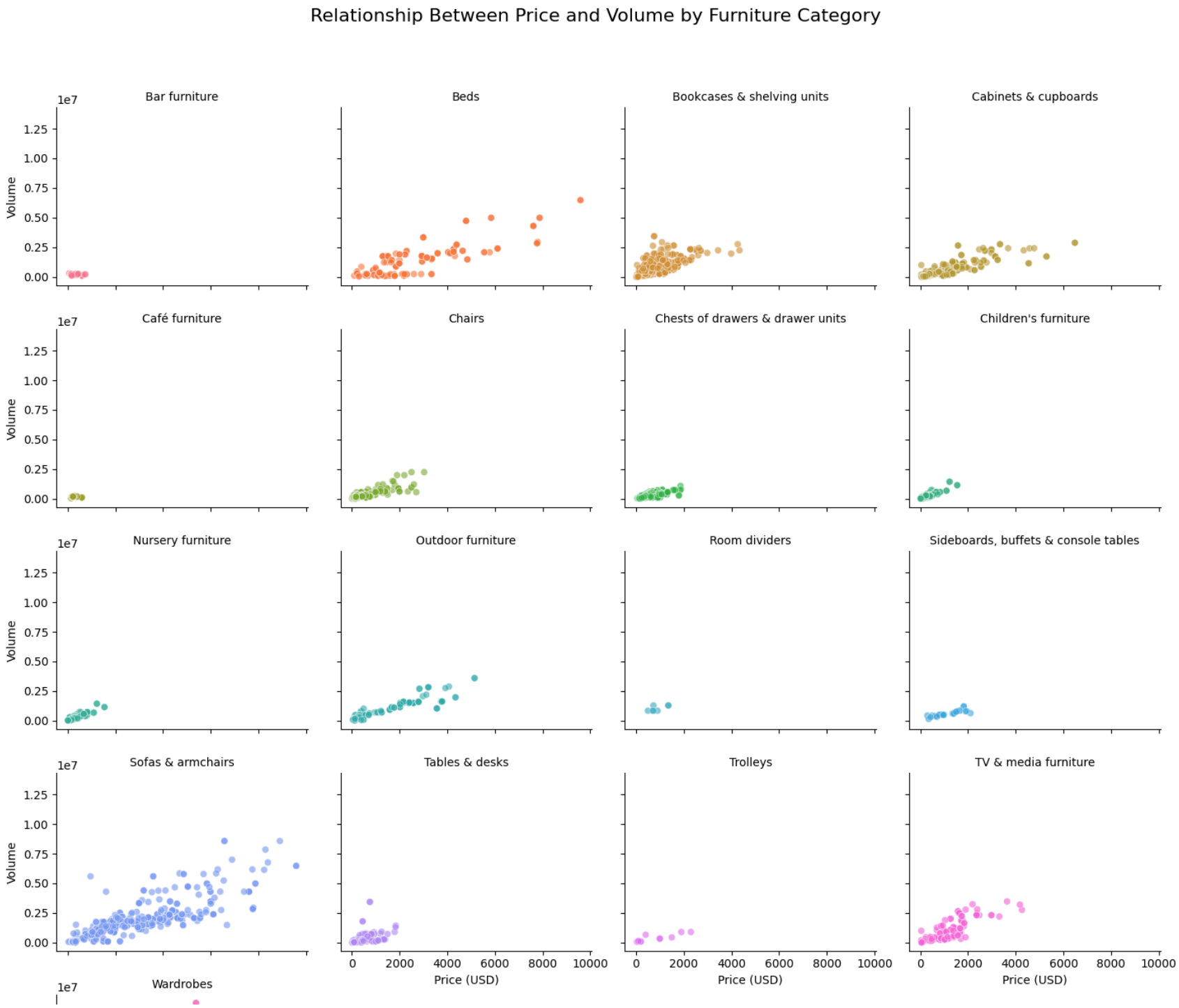
- **2. Key trends:** General Positive Correlation: There is a "cone-shaped" distribution. At lower prices (0–500), volume is generally low. As you move toward the $2,000+ range, the items generally occupy a higher volume (mostly pink dots representing Sofas & armchairs).
- **3. Improvement:** With 17 categories, the legend is overwhelming. It would be much more effective to create 17 small sub-plots (one for each category). This would allow us to see the specific price-to-volume relationship for "Beds" vs. "Chairs" without the dots overlapping.
- 1e7 means 10M

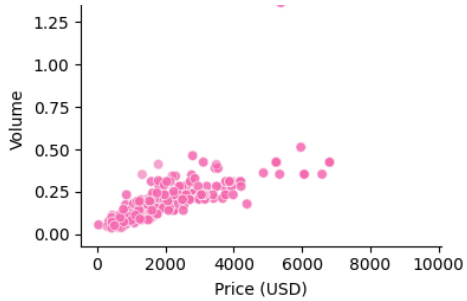
In [14]:

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt


# Calculate Volume for items
data['volume'] = data['depth'] * data['height'] * data['width']

# Create scatter plot
g = sns.relplot(
    data=data,
    x="price", y="volume",
    col="category", hue="category",
    col_wrap=4,        # Number of columns in the grid
    height=3, aspect=1.2,
    kind="scatter", alpha=0.6, legend=False
)

# Improve labels and add a main title
g.set_axis_labels("Price (USD)", "Volume")
g.set_titles("{col_name}")
plt.subplots_adjust(top=0.9)
g.fig.suptitle("Relationship Between Price and Volume by Furniture Category", fontsize=16)

# Save the plot
# plt.savefig('faceted_price_volume_plot.png', bbox_inches='tight')
```

Out[14]:

```
Text(0.5, 0.98, 'Relationship Between Price and Volume by Furniture Category')
```

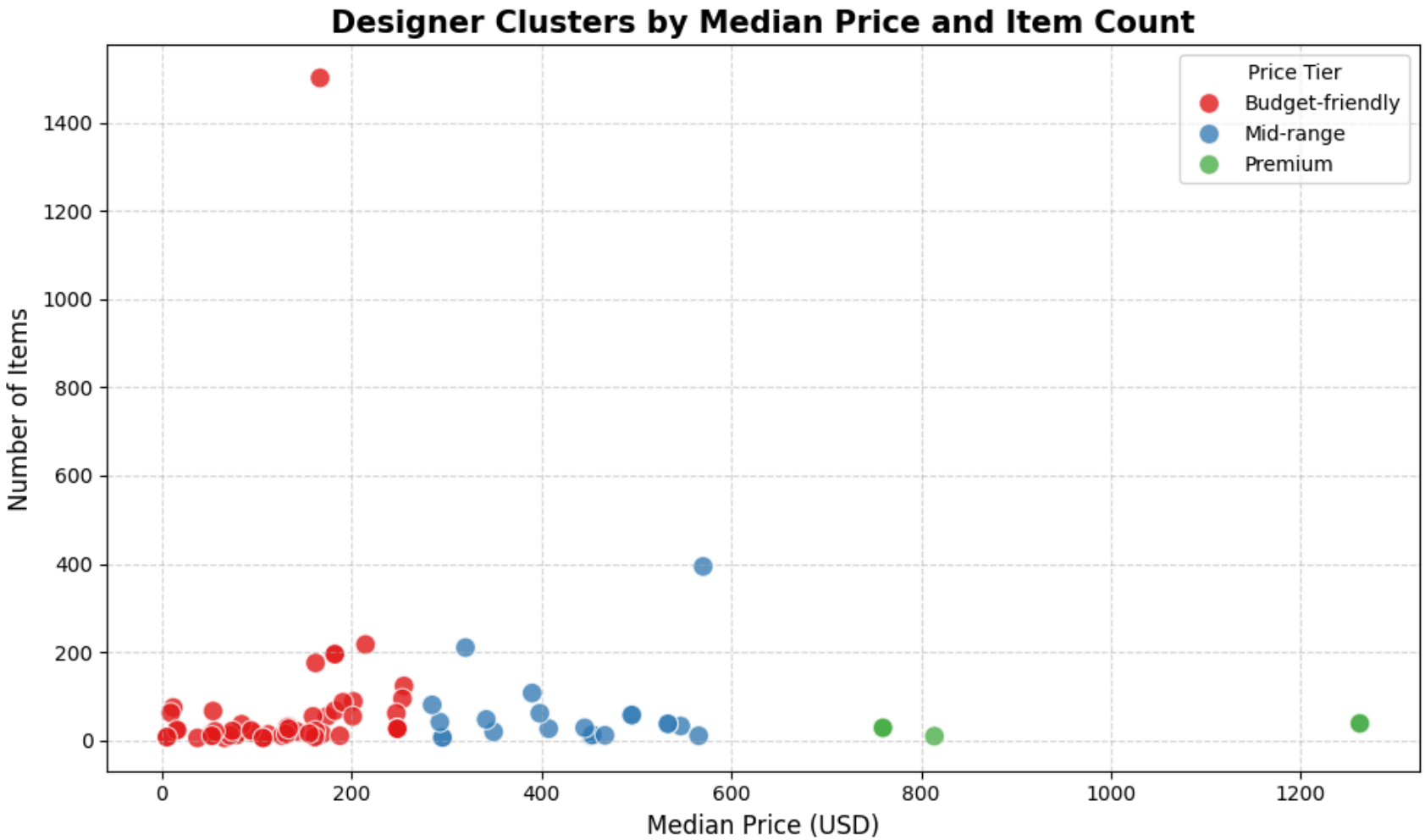

Relationship Between Price and Volume by Furniture Category
Bar furniture
Beds
Bookcases & shelving units
Cabinets & cupboards
Café furniture
Chairs
Chests of drawers & drawer units
Children's furniture
Nursery furniture
Outdoor furniture
Room dividers
Sideboards, buffets & console tables
Sofas & armchairs
Tables & desks
Trolleys
TV & media furniture
Wardrobes
1e7
Volume
0.00
0.25
0.50
0.75
1.00
1.25
0
2000
4000
6000
8000
10000
Price (USD)



Bonus task: Clustering designers by pricing patterns using K-Means clustering

In [17]:
```
from sklearn.cluster import KMeans
import numpy as np

# Step 1: Prepare data - aggregate price data per designer
designer_stats = data.groupby('designer').agg(
    median_price=('price_usd', 'median'),
    std_price=('price_usd', 'std'),
    item_count=('designer', 'count')
).dropna()

# Filter out designers with very few items to reduce noise (e.g., fewer than 5 items)
designer_stats = designer_stats[designer_stats['item_count'] >= 5]

# Step 2: Extract features for clustering (using median price)
# We reshape since sklearn expects a 2D array for features
X = designer_stats['median_price'].values.reshape(-1, 1)

# Step 3: Apply K-Means clustering
# We specify n_clusters=3 (e.g., Budget, Mid-range, Premium)
kmeans = KMeans(n_clusters=3, random_state=42, n_init=10)
designer_stats['cluster'] = kmeans.fit_predict(X)

# Step 4: Visualize the clusters
plt.figure(figsize=(10, 6))

# Mapping clusters to meaningful names based on their sorted cluster centers
centers = kmeans.cluster_centers_.flatten()
```

```
sorted_indices = np.argsort(centers)
cluster_mapping = {sorted_indices[0]: 'Budget-friendly', sorted_indices[1]: 'Mid-range', sorted_indices[2]: 'Premium
designer_stats['cluster_name'] = designer_stats['cluster'].map(cluster_mapping)

sns.scatterplot(
    data=designer_stats,
    x='median_price',
    y='item_count',
    hue='cluster_name',
    palette='Set1',
    s=100,
    alpha=0.8
)

plt.title('Designer Clusters by Median Price and Item Count', fontsize=15, fontweight='bold')
plt.xlabel('Median Price (USD)', fontsize=12)
plt.ylabel('Number of Items', fontsize=12)
plt.legend(title='Price Tier')
plt.grid(True, alpha=0.5, linestyle='--')
plt.tight_layout()
plt.show()

# Display the top designers in the 'Premium' cluster as an example
print("Premium Designers:")
print(designer_stats[designer_stats['cluster_name'] == 'Premium'].sort_values('median_price', ascending=False).head(
```




```
Premium Designers:
              median_price   std_price  item_count  cluster cluster_name
designer
S Lanneskog        1262.25  329.879673          38        1      Premium
J Marnell          1262.25  329.879673          38        1      Premium
E Strandmark        814.05  431.710450           9        1      Premium
Fredriksson         759.78  574.791687          28        1      Premium
Hilland             759.78  574.791687          28        1      Premium
```

By specifying three levels of designers (Budget-friendly, Mid-range and Premium), we can see the relationship between median price and number of itemrs. Specifically, the lower the level of the designers, the more popular their items are for the customers.

What is interesting is that one designer have their products sold the most overall (~1550 items) in the budget-friendly range.

Summary

- The overall distribution of IKEA's product skew to the group of essential furniture (as can be expected) such as "Table & desks" and "Chairs" outweigh others by a relatively high margin. When examining the top product designers, the product tier can be differentiated. Specifically, most designers with popular items (like Francis and Ola) maintain median prices between 200 USD to 300 USD while a higher tier designer (such as Ehlen) main tain the median price of around 600 USD.
- There is a remarkable insight when exploring the relationship between an item's volume and its price, a bigger one doesn't guarantee itself being more expensive than others. K-Means clustering indicates designer's group by three distinct classes that have been mentioned in the bonus task. Beds, "Sofas & Armchairs" with "Wardrobes" are items with the most diverse price range corresponding to the fact that they migh comes to a variaty of sizes.